Newton's Laws of Motion

Introduction

Newton's laws of motion are three fundamental laws that describe the relationship between a body, the forces acting on it and its motion. They were formulated by the English scientist Sir Isaac Newton (1642 to 1727) and published in 1687 in his famous book "Philosophiae Naturalis Principia Mathematica", often called the Principia. Together with his law of universal gravitation, they form the foundation of classical mechanics and explain the motion of everything from a cricket ball to a planet. Engineers use these laws to design vehicles, buildings, rockets and machines.

Key Concepts: Force, Mass and Inertia

A force is a push or a pull that can change the speed, direction or shape of an object. Force is measured in newtons (N). One newton is the force that gives a mass of one kilogram an acceleration of one metre per second squared. Mass is the amount of matter in an object, measured in kilograms; it measures inertia, the tendency of a body to resist a change in its state of motion. A heavier object has more inertia and is harder to start or stop. Weight is the force of gravity on an object, equal to mass times the acceleration due to gravity (about 9.8 metres per second squared on Earth). Mass stays the same everywhere, but weight changes: an object on the Moon weighs about one-sixth of its weight on Earth.

Newton's First Law: The Law of Inertia

Newton's first law states that a body remains at rest, or continues to move in a straight line at constant speed, unless acted upon by an unbalanced external force. In other words, objects do not change their state of motion by themselves. This is also called the law of inertia. Examples include passengers who are thrown forward when a bus brakes suddenly, because their bodies tend to continue moving; dust that flies out of a carpet when it is beaten; a coin that falls into a glass when the card underneath is flicked away; and a ball rolling for a long time on smooth ice. Seat belts and headrests in cars protect people from the effects of inertia. The first law also explains why a spacecraft far from any gravity can keep moving without using fuel. Before Newton, the Greek philosopher Aristotle believed that a force was always needed to keep an object moving; Galileo Galilei corrected this idea by recognising friction as a force.

Newton's Second Law: Force and Acceleration

The second law states that the acceleration of a body is directly proportional to the net force acting on it and inversely proportional to its mass, and the acceleration is in the direction of the force. The formula for Newton's second law is F = m x a, where F is the net force in newtons, m is the mass in kilograms and a is the acceleration in metres per second squared. For example, if a force of 20 newtons acts on a 4 kg trolley, its acceleration is 20 divided by 4, which equals 5 metres per second squared. If the same force acts on a trolley of 10 kg, the acceleration is only 2 metres per second squared. Newton originally stated the law in terms of momentum: the rate of change of momentum of a body equals the net force on it. Momentum is mass times velocity, measured in kilogram metres per second. This explains why it is harder to stop a heavy truck than a bicycle moving at the same speed, and why a cricket fielder pulls his hands backward while catching a fast ball: increasing the time of the catch reduces the force on the hands.

Newton's Third Law: Action and Reaction

The third law states that for every action there is an equal and opposite reaction. When one body exerts a force on a second body, the second body exerts a force of equal size and opposite direction on the first. These two forces always act on different bodies, so they do not cancel each other. A common everyday example of the third law is walking: your foot pushes backward on the ground, and the ground pushes you forward. Other examples include a swimmer pushing water backward and moving forward, a gun recoiling when a bullet is fired, a balloon flying when air rushes out, a boat moving away when a person jumps off it onto the shore, and the way a rocket moves. A rocket pushes hot exhaust gases downward at high speed, and the gases push the rocket upward. This is how ISRO's PSLV, GSLV and LVM3 rockets lift off from Sriharikota and how Chandrayaan-3 reached the Moon.

Applications

Newton's laws are used to design cars with crumple zones and airbags, which increase the time of impact to reduce the force on passengers. They are used to compute the thrust needed for a rocket, to analyse the stress on bridges, to plan satellite orbits, and to understand sports such as cricket, football and badminton. In India, ISRO engineers rely on them for launch trajectories and orbit calculations.

Limitations and Beyond

Newton's laws work extremely well for everyday speeds and sizes, but they fail at speeds close to the speed of light, where Albert Einstein's special relativity is needed, and for very small particles such as electrons, where quantum mechanics applies. Even so, Newtonian mechanics remains the basis of most engineering.

Quick Facts

- First law: a body stays at rest or in uniform motion unless acted on by an unbalanced force (law of inertia).
- Second law: F = m x a, so force equals mass times acceleration.
- Third law: every action has an equal and opposite reaction; walking and rocket propulsion are examples.
- The unit of force is the newton (N).
- Newton published the three laws in the Principia in 1687.